Chapter 11A. Dermatology Overview

Lesion Morphology, Rash Recognition, Bedside Tests, Biopsy Strategy, and Dangerous Rashes

Purpose. This overview chapter is designed as the first dermatology framework for Medvale. It emphasizes the language of morphology, bedside pattern recognition, and the diagnostic decisions that turn a vague complaint of “rash” into a prioritized differential. Later dermatology chapters can then plug specific diseases into this shared vocabulary: papulosquamous disease, eczema, infections, bullous disease, drug reactions, pigment disorders, hair/nail disorders, and skin cancers.

Source synthesis note. The chapter uses the uploaded Anki dermatology-relevant facts as a coverage guide, including cards on cellulitis, tinea, scabies, psoriasis, lichen planus, rosacea, bullous pemphigoid, pemphigus vulgaris, dermatitis herpetiformis, actinic keratosis, basal cell carcinoma, squamous cell carcinoma, melanoma, secondary syphilis, RMSF, meningococcemia, toxic shock syndrome, pyoderma gangrenosum, acanthosis nigricans, vitiligo, and dermatologic manifestations of systemic disease. The organization and prose are original and optimized for future Graph-RAG extraction.

1. The Core Dermatology Problem: Name the Lesion Before Naming the Disease

Dermatology is pattern medicine, but the pattern only becomes useful after the lesion is described precisely. A rash should not initially be labeled “allergy,” “cellulitis,” or “eczema.” It should first be described by morphology, arrangement, distribution, timing, symptoms, and systemic context. Morphology asks what the individual lesion is: macule, patch, papule, plaque, vesicle, bulla, pustule, wheal, nodule, tumor, erosion, ulcer, scale, crust, fissure, lichenification, atrophy, purpura, petechia, or telangiectasia. Arrangement asks whether lesions are annular, linear, grouped, dermatomal, reticular, targetoid, follicular, or confluent. Distribution asks whether the eruption is flexural, extensor, photoexposed, acral, intertriginous, seborrheic, dependent, mucosal, generalized, unilateral, or limited to palms and soles.

A correct descriptive diagnosis can often narrow the differential more than laboratory testing. For example, a **well-demarcated erythematous plaque with silvery scale on extensor surfaces** strongly suggests psoriasis; a **pruritic annular plaque with peripheral scale and central clearing** suggests tinea corporis; a **pink pearly papule with telangiectasias** suggests basal cell carcinoma; and a **palpable purpura** suggests small-vessel vasculitis until proven otherwise. The same “red rash” could represent benign contact dermatitis, cellulitis, meningococcemia, toxic shock syndrome, necrotizing fasciitis, SJS/TEN, acute generalized exanthematous pustulosis, or anaphylaxis. Therefore, the first task is not memorization; it is triage through morphology.

> ***Graph-RAG anchor.*** *A dermatologic concept node should store: lesion morphology, distribution, symptoms, time course, triggers, bedside tests, biopsy/histology clues, systemic associations, and emergency flags. Edges should connect diseases to morphologic patterns, exposures, pathogens, medications, immune mechanisms, and treatment decisions.*

2. Dermatologic History: Timing, Trigger, Texture, and Terrain

The history of a rash should answer five questions: When did it start? Where did it begin? How did it spread? What symptoms accompany it? What exposures or systemic illnesses make a diagnosis dangerous? Time course is often the most decisive feature. Urticaria can appear and disappear within hours; viral exanthems evolve over days; contact dermatitis often follows a sharply demarcated exposure pattern; dermatophyte infections usually expand over weeks; psoriasis, atopic dermatitis, seborrheic dermatitis, rosacea, vitiligo, and hidradenitis suppurativa tend to relapse chronically; and malignancies evolve slowly but persistently, often with bleeding, ulceration, or change in size or color.

Symptoms also direct the differential. **Pruritus** favors eczema, urticaria, scabies, contact dermatitis, tinea, cholestasis, renal disease, hematologic malignancy, or drug eruption. **Pain** suggests zoster, cellulitis, abscess, necrotizing infection, vasculitis with ischemia, erythema nodosum, pyoderma gangrenosum, or ulcerative disease. **Burning or stinging** may be seen in rosacea, zoster prodrome, irritant dermatitis, or small fiber neuropathic syndromes. **Numbness or anesthesia** around an ulcer or plaque raises concern for leprosy, neuropathic ulceration, or deep tissue necrosis. **Skin tenderness out of proportion** is a surgical emergency until necrotizing fasciitis is excluded.

Exposure history includes medications, new topical products, occupational irritants, plants, metals, animals, travel, sexual exposures, sick contacts, hot tubs, pools, gyms, ticks, outdoor activity, immunosuppression, HIV risk, pregnancy, and recent infection. Several high-yield associations recur throughout dermatology: guttate psoriasis after streptococcal infection, erythema multiforme after HSV, lichen planus with hepatitis C, Kaposi sarcoma and bacillary angiomatosis in immunocompromised patients, dermatitis herpetiformis with celiac disease, acanthosis nigricans with insulin resistance or malignancy, pyoderma gangrenosum with inflammatory bowel disease or hematologic disease, erythema nodosum with sarcoidosis, streptococcal infection, tuberculosis, inflammatory bowel disease, pregnancy, drugs, or endemic fungi.

History domain	High-yield questions	Why it matters
Onset and tempo	Minutes-hours, days, weeks, months, recurrent flares, abrupt vs gradual	Immediate urticaria/anaphylaxis differs from viral exanthem, chronic dermatitis, tinea, psoriasis, or malignancy.
Initial site and spread	Started on face, trunk, extremities, palms/soles, mucosa, genitals, scalp, nails	Many infections and inflammatory disorders have stereotyped distribution and spread.
Symptoms	Pruritus, pain, burning, tenderness, numbness, fever, malaise, arthralgia, mucosal pain	Painful rash plus systemic illness is more dangerous than a stable pruritic rash.
Exposures	Drugs, plants, cosmetics, nickel, work chemicals, animals, ticks, travel, sex, sick contacts	Contact dermatitis, drug reactions, arthropod-borne disease, STI, fungal disease, and infestation depend on exposure clues.
Immune context	HIV, transplant, chemotherapy, steroids, biologics, diabetes, ESRD	Immunocompromised patients have atypical infections, severe zoster, ecthyma gangrenosum, Kaposi, bacillary angiomatosis, and disseminated fungi.
Systemic disease	IBD, celiac disease, SLE, dermatomyositis, sarcoid, liver/renal disease, malignancy	Skin can be the first clue to multisystem disease; the rash may be a diagnostic portal rather than a local problem.

3. Primary Lesions: The Alphabet of Dermatology

Primary lesions are the initial visible manifestation of the disease process. A macule is flat and small; a patch is flat and larger. A papule is raised and small; a plaque is raised and broad. A nodule is deeper and often palpable in the dermis or subcutis. A vesicle is a small fluid-filled lesion; a bulla is a large fluid-filled lesion. A pustule contains purulent material and may be infectious, inflammatory, sterile, or follicular. A wheal is transient dermal edema, classically urticaria. A cyst is an encapsulated cavity. A tumor is a large mass-like growth and does not necessarily mean malignant in morphology language, although persistent enlarging tumors require malignancy evaluation.

Primary lesion	Plain-language definition	Classic associations and pattern recognition
Macule / patch	Flat color change. Macule <1 cm; patch >1 cm.	Viral exanthem, drug eruption, toxic shock syndrome, pityriasis rosea herald patch, vitiligo, melasma, cafe-au-lait patch, port-wine stain.
Papule	Small raised solid lesion, usually <1 cm.	Acne, molluscum, scabies burrows/papules, lichen planus, warts, papular urticaria, granuloma annulare.
Plaque	Raised broad lesion, often formed by coalescent papules.	Psoriasis, eczema/lichenification, tinea corporis, nummular dermatitis, cutaneous T-cell lymphoma, actinic keratosis field change.
Nodule	Deeper solid lesion, often dermal/subcutaneous.	Erythema nodosum, rheumatoid nodules, cysts, dermatofibroma, metastasis, deep fungal infection, panniculitis.
Vesicle	Small clear fluid-filled blister.	HSV, VZV, allergic contact dermatitis, dyshidrotic eczema, hand-foot-mouth disease, early bullous disorders.
Bulla	Large fluid-filled blister.	Bullous pemphigoid, pemphigus vulgaris, SJS/TEN, burns, friction, porphyria cutanea tarda, epidermolysis bullosa.
Pustule	Small pus-filled lesion.	Acne, folliculitis, impetigo, disseminated gonococcal infection, pustular psoriasis, AGEP, candidiasis, tinea incognito variants.
Wheal	Transient edematous plaque; changes location over hours.	Urticaria, allergic reaction, physical urticarias, anaphylaxis when systemic symptoms are present.
Purpura / petechia	Non-blanching blood in skin. Petechiae are pinpoint; purpura larger.	Platelet disorders, meningococcemia, RMSF, vasculitis, DIC, anticoagulants, trauma, senile purpura.

4. Secondary Lesions: What the Patient or Time Has Done to the Primary Rash

Secondary lesions develop from scratching, infection, rupture, healing, chronicity, or treatment. They are diagnostically important because they explain why the original morphology may be hidden. For example, atopic dermatitis may present as excoriations and lichenification rather than a fresh eczematous plaque. HSV may be recognized only after vesicles rupture into erosions. Bullous pemphigoid may show urticarial plaques before tense bullae appear. Impetigo may be recognized by honey-colored crust, whereas scabies may show excoriations more than visible mites.

Secondary lesion	Meaning	Clinical use
Scale	Excess stratum corneum shedding; dry flakes.	Psoriasis, tinea, seborrheic dermatitis, ichthyosis, pityriasis rosea, actinic keratosis.
Crust	Dried serum, blood, or pus.	Honey-colored crust suggests impetigo; hemorrhagic crust may occur after scratching, HSV/VZV, ecthyma, or ulcerative disease.
Erosion	Superficial loss of epidermis; heals without scar.	Ruptured vesicles/bullae, HSV, pemphigus vulgaris, intertrigo, excoriated dermatitis.
Ulcer	Deeper loss into dermis/subcutis; may scar.	Venous stasis ulcer, arterial ulcer, neuropathic ulcer, pyoderma gangrenosum, vasculitis, malignancy, infection.
Excoriation	Linear or punctate scratch marks.	Pruritic disorders: atopic dermatitis, scabies, urticaria, cholestasis, renal disease, psychogenic picking.
Lichenification	Thickened skin with accentuated lines from chronic rubbing.	Chronic eczema, lichen simplex chronicus, chronic atopic dermatitis.
Fissure	Linear crack in epidermis/dermis.	Hand dermatitis, tinea pedis, psoriasis, angular cheilitis.
Atrophy	Thinned skin; visible vessels; fragility.	Steroids, aging, lichen sclerosus, morphea, chronic sun damage.
Eschar	Black necrotic tissue.	Ecthyma gangrenosum, anthrax, rickettsial disease, pressure injury, severe ischemia, necrotizing infection.

5. Arrangement and Distribution: The Rash Map

Distribution can be more diagnostic than lesion type. Extensor plaques suggest psoriasis; flexural lichenified eczema suggests atopic dermatitis in older children/adults; seborrheic areas suggest seborrheic dermatitis; intertriginous involvement suggests candidiasis, inverse psoriasis, erythrasma, or irritant intertrigo; dermatomal grouped vesicles suggest herpes zoster; palmar and plantar rash raises concern for secondary syphilis, RMSF, hand-foot-mouth disease, meningococcemia, drug eruption, erythema multiforme, scabies, or eczema; photoexposed rash suggests lupus, dermatomyositis, porphyria, polymorphous light eruption, drug photosensitivity, or actinic damage.

Distribution or arrangement	Differential diagnosis framework
Annular	Tinea corporis, granuloma annulare, erythema migrans, subacute cutaneous lupus, annular psoriasis, erythema annulare centrifugum.
Targetoid	Erythema multiforme, SJS/TEN spectrum, fixed drug eruption, urticaria multiforme; true targets have three zones.
Linear	Allergic contact dermatitis from plants, excoriations, Koebner phenomenon, scabies burrow, lichen striatus, cutaneous larva migrans.
Dermatomal	Herpes zoster, zoster sine herpete if pain without rash; consider ophthalmic involvement with V1 distribution.
Reticular / livedoid	Livedo reticularis, antiphospholipid syndrome, cholesterol emboli, vasculitis, vasculopathy, cryoglobulinemia, Sneddon syndrome.
Follicular	Folliculitis, keratosis pilaris, acneiform eruption, steroid acne, hot-tub folliculitis, pityrosporum folliculitis.
Intertriginous	Candidal intertrigo, tinea cruris, erythrasma, inverse psoriasis, hidradenitis suppurativa, irritant dermatitis.
Seborrheic areas	Seborrheic dermatitis, rosacea, acne, lupus/photosensitivity mimics, dermatomyositis facial involvement.
Palms/soles	Secondary syphilis, RMSF, hand-foot-mouth disease, erythema multiforme, scabies, drug eruption, meningococcemia, psoriasis, dyshidrotic eczema.
Mucosal	HSV, aphthous ulcers, Behcet, SJS/TEN, pemphigus vulgaris, lichen planus, candidiasis, secondary syphilis, Crohn disease.

Pattern shortcut. *A rash on extensor elbows/knees with scale is a different clinical object than a rash in flexures with lichenification, a dermatomal vesicular eruption, a reticular violaceous eruption, a non-blanching petechial eruption, or a mucosal erosive eruption. In a Graph-RAG system, each distribution acts as an edge activator connecting morphology to disease clusters.*

6. Blanching, Non-Blanching, and Vascular Clues

Blanching is tested by pressing a glass slide or finger on erythematous skin. Blanching suggests vascular dilation; non-blanching suggests blood outside vessels or occlusive/hemorrhagic pathology. Petechiae and purpura should not be dismissed as “rash,” especially when accompanied by fever, hypotension, severe headache, meningismus, thrombocytopenia, renal disease, abdominal pain, neuropathy, or rapidly progressive illness. Palpable purpura implies inflammation of small vessels, whereas flat petechiae may reflect thrombocytopenia, platelet dysfunction, DIC, sepsis, or rickettsial disease.

Vascular patterns connect dermatology to internal medicine. Livedo reticularis and blue toe syndrome may point toward cholesterol embolization or antiphospholipid syndrome. Telangiectasias are prominent in rosacea, systemic sclerosis/CREST, hereditary hemorrhagic telangiectasia, chronic liver disease, and chronic sun damage. Erythema nodosum is a tender panniculitis with nodules on the shins and is a clue to sarcoidosis, streptococcal infection, tuberculosis, inflammatory bowel disease, pregnancy, drugs, or endemic fungal infection such as coccidioidomycosis.

7. Bedside Diagnostic Tests and Office Procedures

Many dermatologic diagnoses are clinical, but bedside tests are high-yield when the differential includes fungi, scabies, HSV/VZV, pigment disorders, erythrasma, contact allergy, or malignancy. Testing should be selected to answer a focused question. A KOH preparation asks, “Is there a dermatophyte or yeast?” A Wood lamp asks, “Is there fluorescence from certain organisms or pigment contrast?” Tzanck smear is historically associated with multinucleated giant cells in herpes infections but is less specific than PCR and is now mainly a classic teaching concept. Dermoscopy helps evaluate pigmented and vascular lesions. Biopsy is appropriate when malignancy, vasculitis, bullous disease, panniculitis, uncertain persistent rash, or dangerous drug eruption is suspected.

Test	What it helps diagnose	Key interpretation / caveat
KOH preparation	Tinea corporis/cruris/pedis, candidiasis, tinea versicolor	Branching septate hyphae suggest dermatophyte; budding yeast/pseudohyphae suggest Candida; short hyphae and spores can suggest Malassezia.
Wood lamp	Erythrasma, some tinea capitis, pigment disorders, porphyria clues	Coral-red fluorescence suggests Corynebacterium erythrasma; pigment contrast may help vitiligo; many dermatophytes do not fluoresce.
Dermoscopy	Melanoma, BCC, SCC, seborrheic keratosis, vascular lesions	Improves pattern recognition but does not replace biopsy when melanoma is suspected.
Skin scraping	Scabies, mites, dermatophytes	Scrape burrows or scale; a negative scraping does not always exclude scabies if clinical suspicion is high.
Bacterial culture	Purulent cellulitis, abscess, impetigo outbreaks, ulcers	Low yield in nonpurulent cellulitis; culture pus, drainage, abscess material, or deep tissue when possible.
Viral PCR	HSV, VZV, mpox when suspected	Preferred over nonspecific cytology for vesicular or ulcerative lesions.
Patch testing	Allergic contact dermatitis	Identifies delayed type IV hypersensitivity triggers such as nickel, fragrances, preservatives, rubber accelerators, or topical medications.
Diascopy	Blanching vascular lesions, granulomatous inflammation	Apple-jelly color can be a clue to granulomatous lesions, but biopsy is often needed.
Biopsy	Cancer, vasculitis, bullous disease, panniculitis, persistent unclear rash	Technique and site matter; wrong lesion or wrong depth can make pathology nondiagnostic.

8. Biopsy Strategy: Match Technique to the Diagnostic Question

Skin biopsy is not one test; it is a family of sampling strategies. The best biopsy depends on lesion depth, suspected disease, and pathology needs. For suspected melanoma, the preferred initial procedure is generally a full-thickness excisional biopsy with narrow margins when feasible, because Breslow depth is the most important staging/prognostic feature and partial sampling can underestimate invasion. For inflammatory rashes, punch biopsy is often useful because it samples epidermis, dermis, and superficial subcutis. For vesiculobullous disease, direct immunofluorescence usually requires perilesional normal-appearing skin rather than the eroded center of a blister. For vasculitis, a fresh purpuric lesion is preferred; older lesions may show nonspecific changes. For panniculitis, a deep incisional or large punch biopsy that reaches subcutaneous fat may be required.

Clinical question	Preferred sample	Pitfall to avoid
Suspected melanoma	Full-thickness excisional biopsy with narrow clinical margins when anatomically feasible	Do not shave transect a suspicious pigmented lesion if depth assessment may be compromised.
Nonmelanoma skin cancer	Shave, saucerization, punch, or excision depending on lesion and site	Superficial sampling may miss invasive SCC in a hyperkeratotic lesion.
Inflammatory rash	Punch biopsy of representative active lesion	Avoid excoriated, infected, or steroid-treated areas if possible.
Bullous disease	Two biopsies: lesional for H&E; and perilesional for direct immunofluorescence	Do not send only ruptured erosions; immunofluorescence site matters.
Vasculitis	Fresh palpable purpura, often punch including dermis; DIF can be added	Old lesions may have burned-out nonspecific inflammation.
Panniculitis/nodules	Deep punch/incisional biopsy including subcutaneous fat	Too shallow biopsy misses the diagnostic compartment.
Ulcer	Biopsy edge and sometimes base if malignancy/infection/vasculitis considered	Do not sample only necrotic center.
Alopecia	Scalp punch from active margin; orientation and horizontal sections may be needed	Wrong site can miss scarring vs nonscarring diagnosis.

9. The Rash Algorithm: Triage Before Diagnosis

A practical rash algorithm begins with danger. Determine whether the patient is unstable, febrile, toxic-appearing, immunocompromised, has mucosal involvement, skin pain, non-blanching lesions, necrosis, rapidly spreading erythema, bullae, facial/airway swelling, ocular symptoms, or concern for abuse/infestation outbreak. If any of these are present, the rash becomes an emergency problem rather than an outpatient pattern-recognition exercise.

Step 1 - Stability: hypotension, altered mental status, respiratory symptoms, angioedema, fever with petechiae/purpura, or severe pain = urgent evaluation.

Step 2 - Morphology: macular/papular, papulosquamous, vesicular/bullous, pustular, urticarial, purpuric, ulcerative, nodular, pigmentary, or hair/nail.

Step 3 - Distribution: localized vs generalized; dermatomal vs symmetric; flexural vs extensor; intertriginous vs seborrheic; photoexposed vs covered; palms/soles/mucosa/genitals.

Step 4 - Timing: minutes-hours suggests urticaria/anaphylaxis; days suggests infection/drug/viral exanthem; weeks-months suggests dermatitis/tinea/psoriasis; persistent growth suggests neoplasm.

Step 5 - Tests: KOH for scale/annular lesions; HSV/VZV PCR for vesicles/ulcers; culture purulence; biopsy suspicious pigmented, bullous, vasculitic, ulcerative, nodular, or persistent unexplained lesions.

Step 6 - Initial treatment: avoid steroids on undiagnosed fungal infection, avoid missing cellulitis/necrotizing infection, avoid delayed biopsy of melanoma, and protect mucosa/eyes in severe drug reactions.

> ***Teaching trap.*** *Topical steroids can temporarily reduce inflammation in tinea, creating tinea incognito. Any annular scaly rash that worsens or spreads with steroids should trigger KOH examination or empiric antifungal strategy rather than escalating steroid potency.*

10. Dangerous Rashes: Red Flags That Change the Setting of Care

Dangerous rashes are defined by speed, pain, systemic toxicity, mucosal involvement, vascular compromise, infection risk, or malignant potential. These conditions may initially resemble benign eruptions, but the consequences of delay are high. A patient with fever and petechiae may have meningococcemia or RMSF. A patient with skin pain and dusky erythema may have necrotizing fasciitis or early SJS/TEN. A patient with urticaria plus respiratory symptoms or hypotension has anaphylaxis. A patient with a new changing pigmented lesion requires biopsy strategy, not reassurance. A patient with ophthalmic zoster may lose vision. A patient with purpura and renal disease may have vasculitis. A patient with immunosuppression and necrotic lesions may have ecthyma gangrenosum, disseminated fungal disease, or invasive infection.

Red flag	Why it matters	Examples
Fever + petechiae/purpura	Sepsis, meningitis, rickettsial disease, DIC, vasculitis	Meningococcemia, RMSF, severe thrombocytopenia, leukocytoclastic vasculitis.
Pain out of proportion	Deep tissue necrosis may precede skin findings	Necrotizing fasciitis, compartment ischemia, severe cellulitis, vasculopathy.
Mucosal erosions + skin pain	Severe drug reaction/bullous disease risk	SJS/TEN, erythema multiforme major, pemphigus vulgaris.
Rapidly spreading erythema/systemic toxicity	Invasive infection or toxin-mediated disease	Cellulitis, erysipelas, necrotizing infection, toxic shock syndrome.
Bullae, dusky skin, necrosis, eschar	Vascular compromise, necrosis, severe infection/drug reaction	Necrotizing fasciitis, ecthyma gangrenosum, anthrax, SJS/TEN, vasculitis.
Facial swelling/airway symptoms	Anaphylaxis or angioedema	Urticaria with wheeze/hypotension, ACE-inhibitor angioedema.
Ocular pain/redness with rash	Vision-threatening disease	Herpes zoster ophthalmicus, SJS/TEN, gonococcal/chlamydial disease, Behcet.
New/changing pigmented lesion	Melanoma must be excluded	ABCDE change, ugly duckling, bleeding/itching/nodularity, acral lesion.
Immunocompromised host	Atypical organisms and disseminated infection	Disseminated zoster, ecthyma gangrenosum, Kaposi, bacillary angiomatosis, fungal infection.
Palpable purpura + renal/neurologic/GI findings	Systemic vasculitis until proven otherwise	IgA vasculitis, ANCA vasculitis, cryoglobulinemia, drug-induced vasculitis.

11. Papulosquamous and Eczematous Pattern Recognition

Papulosquamous diseases are raised erythematous lesions with scale. The core differential includes psoriasis, tinea, pityriasis rosea, lichen planus, seborrheic dermatitis, secondary syphilis, and cutaneous T-cell lymphoma. Eczematous diseases are inflammatory, itchy, often poorly demarcated, and may be acute with vesiculation or chronic with lichenification. The core differential includes atopic dermatitis, allergic contact dermatitis, irritant contact dermatitis, nummular eczema, dyshidrotic eczema, stasis dermatitis, and seborrheic dermatitis overlap.

Pattern	Clues	Initial diagnostic/treatment thought
Psoriasis	Well-demarcated plaques, silvery scale, extensor surfaces, scalp/nails, Koebner phenomenon; guttate after strep/HIV	Clinical diagnosis; assess joints/nails; topical steroids/vitamin D analogs for limited disease; systemic/biologic therapy for severe disease.
Tinea corporis	Annular scaly plaque with central clearing; pruritic; may worsen with steroids	KOH if uncertain; topical azoles/allylamines for localized disease.
Pityriasis rosea	Herald patch followed by Christmas-tree trunk eruption	Usually self-limited; consider syphilis testing if palms/soles/mucosa or risk factors.
Lichen planus	Pruritic purple polygonal planar papules/plaques; Wickham striae; wrists/ankles/oral/genital; hepatitis C association	Topical steroids; biopsy if uncertain or mucosal disease.
Seborrheic dermatitis	Greasy yellow scale in scalp, eyebrows, nasolabial folds, chest; dandruff	Antifungal shampoos/creams, low-potency steroids briefly for inflammation.
Atopic dermatitis	Chronic relapsing pruritus; flexural lichenification in older children/adults; barrier dysfunction	Emollients, trigger avoidance, topical anti-inflammatory therapy; treat infection if impetiginized.
Allergic contact dermatitis	Pruritic, often sharply geometric or linear; vesicles after poison ivy; nickel common	Avoid trigger; topical steroids; patch testing for recurrent unclear disease.
Stasis dermatitis	Lower legs with edema, hemosiderin, varicosities, pruritus/scale; may ulcerate	Compression/edema control; avoid misdiagnosing bilateral stasis as cellulitis.

12. Vesicular, Bullous, and Erosive Patterns

Vesicles and bullae are high-yield because they force a split between infection, autoimmune blistering disease, drug reaction, contact dermatitis, and mechanical injury. Grouped vesicles on an erythematous base suggest HSV; dermatomal vesicles with neuropathic pain suggest zoster; diffuse painful mucosal erosions with flaccid bullae suggest pemphigus vulgaris; tense bullae in an older patient with pruritic urticarial plaques suggest bullous pemphigoid; intensely pruritic grouped vesicles on extensor surfaces suggest dermatitis herpetiformis; widespread targetoid lesions after HSV suggest erythema multiforme; skin pain, mucosal involvement, and epidermal detachment suggest SJS/TEN.

Disease/pattern	Morphology	Mechanism / distinguishing clue
HSV	Painful grouped vesicles on erythematous base; oral/genital/whitlow	PCR from lesion if uncertain; recurrent; herpetic whitlow in healthcare/dental exposure.
VZV/zoster	Dermatomal vesicles with pain/burning prodrome	V1 distribution or eye symptoms = urgent ophthalmology consideration.
Pemphigus vulgaris	Flaccid bullae/erosions, mucosal involvement, positive Nikolsky	Autoantibodies to desmogleins; intercellular “fish-net” immunofluorescence.
Bullous pemphigoid	Tense bullae, older adults, pruritic urticarial plaques, mucosa less common	Autoantibodies at basement membrane; linear immunofluorescence; eosinophils.
Dermatitis herpetiformis	Grouped intensely pruritic vesicles/papules on extensor elbows/knees/buttocks	Celiac association; granular IgA at dermal papillae; responds to gluten-free diet/dapsone.
Contact dermatitis	Linear/geometric vesicles, intense pruritus, exposure pattern	Poison ivy and nickel are classic; treat with avoidance and anti-inflammatory therapy.
SJS/TEN	Painful dusky/targetoid lesions, mucosal erosions, epidermal sloughing	Medication/infection trigger; inpatient/burn-unit style supportive care; stop culprit drug.
Erythema multiforme	Target lesions often after HSV; may have limited mucosal disease	Different from SJS/TEN by classic targets and less sheet-like necrosis.

13. Infectious and Infestation Clusters in a Dermatology Overview

The infectious rash differential is best organized by morphology and host. Bacterial infections include impetigo with honey-colored crust, ecthyma with deeper ulcerative crusts, cellulitis with poorly demarcated erythematous tender deep dermal/subcutaneous infection, erysipelas with sharply demarcated superficial infection, abscess with fluctuance, folliculitis with follicular pustules, necrotizing fasciitis with severe pain and systemic toxicity, and toxin-mediated syndromes such as toxic shock syndrome with diffuse erythematous macular rash and shock. Viral patterns include HSV, VZV, parvovirus B19, measles/rubella/roseola patterns, enterovirus hand-foot-mouth disease, and mpox when epidemiology and lesion morphology fit. Fungal patterns include tinea, candidal intertrigo, tinea versicolor, and endemic fungi with skin involvement. Infestations include scabies with nocturnal pruritus, burrows, finger web/genital/wrist involvement, and household spread.

Condition	Morphologic clue	Internal medicine relevance
Cellulitis	Warm tender erythema with indistinct borders; usually unilateral	Deep dermis/subcutaneous infection; bilateral lower leg erythema often suggests stasis dermatitis rather than cellulitis.
Impetigo	Honey-colored crust, often face/extremities	S. aureus or GAS; bullous form often S. aureus toxin.
Necrotizing fasciitis	Pain out of proportion, rapid progression, bullae/necrosis, systemic toxicity	Surgical emergency; imaging must not delay operative management when suspicion is high.
Ecthyma gangrenosum	Rapid necrotic eschar with surrounding erythema, often immunocompromised/neutropenic	Classically Pseudomonas bacteremia but differential includes other invasive infections.
Tinea	Annular scale, central clearing; KOH positive	Avoid steroid monotherapy; tinea cruris can recur from untreated tinea pedis autoinfection.
Scabies	Nocturnal pruritus, burrows, finger webs/wrists/genitals, household contacts	Treat patient and contacts; permethrin first-line in many settings; ivermectin alternative/special cases.
Secondary syphilis	Diffuse maculopapular rash, palms/soles, mucous patches, lymphadenopathy	Do not miss in generalized rash; test based on risk and distribution.
RMSF	Fever/headache with rash from wrists/ankles to trunk/palms/soles; petechial later	Treat empirically with doxycycline when suspected; labs may show thrombocytopenia/transaminitis.
Meningococcemia	Fever, shock, petechiae/purpura, cool mottled skin	Immediate antibiotics and sepsis management.
Disseminated gonococcal infection	Migratory polyarthralgia/tenosynovitis with pustular rash	Sexual history, NAAT/cultures, ceftriaxone-based therapy.

14. Pigmentary Lesions and Skin Cancer Screening Logic

Pigmented lesions require a structured malignancy screen. The ABCDE heuristic asks whether a lesion is asymmetric, has irregular borders, color variation, diameter commonly greater than 6 mm, or evolution. The “ugly duckling” sign asks whether one lesion looks different from the patient’s other nevi. Symptoms such as bleeding, itching, pain, ulceration, sensory change, or palpable nodularity increase concern. The most important prognostic factor for melanoma metastasis is Breslow depth, which is why the initial biopsy should preserve depth assessment. Acral lentiginous melanoma is an important subtype on palms, soles, and nail apparatus and is the most common melanoma subtype in many darker-skinned populations, so low overall melanoma incidence must not lead to dismissal of acral lesions.

Lesion	Classic description	High-yield action
Actinic keratosis	Rough/flaky sandpaper-like lesion on sun-exposed skin; SCC precursor	Cryotherapy for limited lesions; field therapy such as 5-FU/imiquimod for diffuse field disease.
Basal cell carcinoma	Pink pearly papule with telangiectasias; rolled border; may ulcerate	Biopsy suspicious lesion; local destructive/excisional therapy depending on risk and site.
Squamous cell carcinoma	Ulcerative or hyperkeratotic red lesion with scale; may arise in scars/chronic wounds/immunosuppression	Biopsy; higher metastatic potential than BCC; chronic scars and immunocompromise are important risk settings.
Melanoma	Changing pigmented lesion; ABCDE or ugly duckling; nodularity/bleeding/itching concerning	Full-thickness excisional biopsy with narrow margins when feasible; Breslow depth drives staging.
Seborrheic keratosis	Waxy stuck-on plaque; can be pigmented and verrucous	Benign, but biopsy if atypical or cannot distinguish from melanoma.
Acanthosis nigricans	Velvety hyperpigmented plaques in neck/axilla/groin	Usually insulin resistance; abrupt extensive disease can be paraneoplastic.
Vitiligo	Depigmented patches from autoimmune melanocyte destruction	Associated with autoimmune disease; Wood lamp can accentuate depigmentation.
Tinea versicolor	Hypo- or hyperpigmented trunk patches after heat/humidity; fine scale	Malassezia; KOH may show short hyphae/spores; recurrence common.

15. Hair, Nail, Mucosal, and Genital Clues

Dermatology is not limited to visible skin. Nails can indicate psoriasis, lichen planus, onychomycosis, iron deficiency, systemic disease, trauma, melanoma, or endocarditis. Nail pitting, onycholysis, and oil-drop changes support psoriasis and psoriatic arthritis. Oral mucosal disease suggests HSV, aphthous ulcers, Behcet disease, pemphigus vulgaris, lichen planus, candidiasis, secondary syphilis, inflammatory bowel disease, or SJS/TEN. Genital lesions require STI, inflammatory, neoplastic, and autoimmune consideration; painful vesicles on an erythematous base suggest HSV, painless chancre suggests primary syphilis, and penile lichen sclerosus may require biopsy when suspected.

Hair loss should first be split into scarring versus nonscarring alopecia. Scarring alopecia destroys follicles and requires prompt dermatology evaluation because regrowth may not occur. Nonscarring alopecia includes androgenetic alopecia, telogen effluvium, alopecia areata, traction alopecia, tinea capitis, thyroid disease, iron deficiency, medications, and systemic illness. Alopecia with rash can suggest autoimmune disease, syphilis, zinc deficiency, severe seborrheic dermatitis, tinea capitis, or drug reactions.

16. Systemic Disease Written on the Skin

Skin findings often serve as external signs of internal medicine. Dermatomyositis presents with heliotrope rash and Gottron papules in association with proximal muscle weakness and malignancy risk. Systemic sclerosis causes skin thickening, sclerodactyly, telangiectasias, Raynaud phenomenon, and salt-and-pepper pigment changes. SLE can cause photosensitive malar or discoid rashes, oral ulcers, alopecia, and vasculitic lesions. Inflammatory bowel disease is associated with erythema nodosum and pyoderma gangrenosum. Celiac disease is associated with dermatitis herpetiformis. Diabetes predisposes to candidiasis, acanthosis nigricans, necrobiosis lipoidica, ulcers, cellulitis, and severe infection. ESRD can produce pruritus and calciphylaxis with painful ischemic skin necrosis. Liver disease may produce jaundice, pruritus, spider angiomas, palmar erythema, and xanthomas. Hemochromatosis classically links bronze skin, diabetes, and cirrhosis.

Skin finding	Internal association	Do not miss
Gottron papules / heliotrope rash	Dermatomyositis	Check muscle weakness, CK/aldolase, ILD symptoms, malignancy context.
Palpable purpura	Small-vessel vasculitis, infection, drug, cryoglobulins, IgA vasculitis	Urinalysis/renal function and systemic symptoms determine urgency.
Erythema nodosum	Sarcoid, strep, TB, IBD, pregnancy, drugs, endemic fungi	Tender shin nodules are panniculitis, not cellulitis.
Pyoderma gangrenosum	IBD, arthritis, hematologic disease	Pathergy; debridement can worsen; diagnose after excluding infection.
Acanthosis nigricans	Insulin resistance; rarely gastric malignancy/paraneoplastic	Abrupt widespread disease or mucosal involvement is concerning.
Dermatitis herpetiformis	Celiac disease	Grouped pruritic extensor vesicles; test and treat systemic disease.
Livedo/blue toe	APS, cholesterol emboli, vasculopathy, vasculitis	Renal dysfunction/eosinophilia after vascular procedure suggests cholesterol emboli.
Calciphylaxis	ESRD, warfarin, calcium-phosphate abnormalities	Painful retiform purpura/necrosis; high mortality and infection risk.

17. Initial Treatment Principles Without Over-Treating the Wrong Disease

A dermatology overview should emphasize safe first steps rather than memorizing every drug. Moisturizers and barrier repair are foundational for eczema and xerosis. Topical corticosteroids reduce inflammatory dermatitis, psoriasis plaques, lichen simplex chronicus, and many pruritic inflammatory eruptions, but potency must match body site and duration. High-potency steroids are generally avoided on the face, groin, and intertriginous areas unless directed by dermatology. Topical antifungals treat localized tinea and candidiasis; oral therapy is required for many scalp, nail, extensive, or refractory fungal infections. Antibiotics are used when bacterial infection is likely, especially cellulitis, impetigo, abscess with systemic features, or high-risk purulent infection. Antihistamines help urticaria and allergic pruritus but do not treat anaphylaxis, vasculitis, or severe drug reactions.

The safest rule is to avoid using topical steroids as a diagnostic shortcut when fungal infection, scabies, bacterial infection, HSV, or malignancy is plausible. Steroids can mask infection, worsen tinea, delay diagnosis of cancer, and thin skin when overused. Conversely, undertreating true inflammatory skin disease can lead to chronic lichenification, sleep disruption, secondary infection, and impaired quality of life. The balance is achieved through morphology, focused testing, and follow-up.

Therapy class	Useful for	Major cautions
Emollients/barrier repair	Atopic dermatitis, xerosis, irritant dermatitis, chronic pruritus support	Requires frequent use; ointments often better barrier but greasier.
Topical corticosteroids	Eczema, psoriasis plaques, contact dermatitis, lichen simplex, inflammatory flares	Atrophy, striae, telangiectasias, perioral dermatitis, glaucoma risk near eyes; avoid masking infection.
Topical calcineurin inhibitors	Face/flexural atopic dermatitis, steroid-sparing therapy	Burning sensation; useful where steroid atrophy is a concern.
Topical antifungals	Localized tinea corporis/cruris/pedis, candidal intertrigo	KOH if uncertain; avoid steroid-antifungal combinations as reflex therapy.
Systemic antifungals	Tinea capitis, onychomycosis, extensive/refractory disease	Drug interactions, liver considerations, organism/site-specific choices.
Antibacterial therapy	Cellulitis, impetigo, ecthyma, selected abscess cases	Culture purulence when possible; drainage is key for abscess.
Antihistamines	Urticaria/pruritus symptom control	Anaphylaxis requires epinephrine; sedating agents impair driving/alertness.
Biologics/systemics	Moderate-severe psoriasis, atopic dermatitis, hidradenitis, autoimmune disease	Screening and infection risk depend on agent; specialty-managed.

18. High-Yield Mini-Differentials for Fast Recall

Presentation	Prioritized differential
Itchy annular plaque with scale	Tinea corporis, nummular eczema, psoriasis, granuloma annulare, subacute cutaneous lupus.
Diffuse maculopapular rash with palms/soles	Secondary syphilis, RMSF, drug eruption, meningococcemia, viral exanthem, hand-foot-mouth disease.
Grouped vesicles	HSV if localized oral/genital/whitlow; zoster if dermatomal; dermatitis herpetiformis if extensor and intensely pruritic.
Tense bullae in older adult	Bullous pemphigoid, drug-induced bullous eruption, edema/friction bullae, porphyria cutanea tarda.
Flaccid bullae/erosions + mucosa	Pemphigus vulgaris, SJS/TEN, severe HSV, erosive lichen planus, Behcet.
Painful purple ulcer with undermined border	Pyoderma gangrenosum, vasculitis, infection, malignancy, arterial ulcer.
Tender shin nodules	Erythema nodosum from sarcoid, strep, TB, IBD, pregnancy, drugs, endemic fungi.
Velvety hyperpigmented neck/axilla	Acanthosis nigricans from insulin resistance; abrupt extensive disease suggests malignancy workup.
Pink pearly papule with telangiectasia	Basal cell carcinoma until proven otherwise.
Hyperkeratotic scaly ulcerated plaque	Squamous cell carcinoma or actinic keratosis spectrum; biopsy suspicious lesions.
Changing pigmented lesion	Melanoma evaluation with excisional biopsy when feasible.
Skin pain + dusky lesions + mucosa	SJS/TEN or necrotizing infection depending morphology/systemic signs; urgent care.

19. Dermatology Overview Diagnostic Algorithms

Algorithm A: Generalized Rash

1. Check vital signs and toxicity. If unstable, febrile with petechiae/purpura, mucosal sloughing, severe pain, or immunocompromised with necrosis, escalate care.
2. Identify morphology: maculopapular vs papulosquamous vs urticarial vs vesicular/bullous vs pustular vs purpuric vs nodular/ulcerative.
3. Ask about new medications in the prior days-weeks. Drug eruption, urticaria, DRESS, AGEP, and SJS/TEN are time-linked diagnoses.
4. Examine palms, soles, mouth, eyes, genitals, scalp, and nails. Many missed diagnoses are found off the trunk.
5. Use targeted testing: syphilis/HIV when risk or palms/soles/mucosa; CBC/CMP/UA if systemic disease; KOH for scale; PCR for vesicles/ulcers; biopsy for vasculitis/bullous/neoplasm/unclear persistent disease.

Algorithm B: Scaly Plaque

1. Is it annular with leading scale and central clearing? Perform KOH or treat as tinea if classic.
2. Is it well-demarcated on extensor surfaces/scalp/nails? Think psoriasis and assess for joint symptoms.
3. Is it greasy yellow scale in seborrheic areas? Think seborrheic dermatitis.
4. Is it flexural, intensely pruritic, chronic, and lichenified? Think atopic dermatitis or chronic eczema.
5. Is it purple, polygonal, planar, pruritic with oral/genital involvement? Think lichen planus.

6. Does it persist, atrophy, ulcerate, bleed, or resist therapy? Biopsy to exclude CTCL, SCC, or other malignancy.

Algorithm C: Blistering or Erosive Rash

1. Are lesions localized grouped vesicles? Consider HSV; dermatomal pain/vesicles suggests zoster.
2. Are bullae tense in an older adult with pruritus and minimal mucosa? Think bullous pemphigoid.
3. Are bullae flaccid with mucosal erosions and positive Nikolsky? Think pemphigus vulgaris.
4. Are there painful dusky lesions, mucosal involvement, and epidermal detachment? Treat as SJS/TEN emergency.
5. For suspected autoimmune blistering disease, biopsy both lesional tissue for H&E; and perilesional skin for direct immunofluorescence.

Algorithm D: Pigmented Lesion

1. Compare with patient’s other lesions: ugly duckling is high risk.
2. Apply ABCDE: asymmetry, border irregularity, color variation, diameter, evolution.
3. Ask about bleeding, itching, pain, sensory change, ulceration, or palpable nodularity.
4. Examine acral/nail lesions carefully, especially in darker skin where acral lentiginous melanoma is disproportionately important.
5. If suspicious, perform or refer for full-thickness excisional biopsy with narrow margins when feasible; do not reassure based only on young age.

20. Chapter 11A Summary Schema for Graph-RAG

Node/edge type	Examples to encode
Morphology nodes	Macule, patch, papule, plaque, vesicle, bulla, pustule, wheal, nodule, purpura, ulcer, scale, crust, lichenification.
Distribution nodes	Extensor, flexural, seborrheic, intertriginous, dermatomal, photoexposed, palmar/plantar, mucosal, acral, generalized.
Disease cluster nodes	Papulosquamous, eczematous, infectious, infestation, vesiculobullous, purpuric/vasculitic, pigmentary, neoplastic, hair/nail, mucosal/genital.
Diagnostic test nodes	KOH, Wood lamp, scraping, dermoscopy, PCR, culture, patch testing, biopsy, direct immunofluorescence.
Emergency edges	Fever+petechiae -> meningococcemia/RMSF/sepsis; skin pain+mucosa -> SJS/TEN; pain out of proportion -> necrotizing infection; changing pigment -> melanoma biopsy.
Systemic association edges	Dermatitis herpetiformis-celiac; lichen planus-HCV; erythema nodosum-sarcoid/IBD/TB; pyoderma gangrenosum-IBD; acanthosis-insulin resistance/malignancy; calciphylaxis-ESRD.
Treatment caution edges	Steroids can worsen tinea; abscess requires drainage; anaphylaxis requires epinephrine; suspicious pigmented lesion requires biopsy; severe drug rash requires drug cessation and urgent care.

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Uploaded Medvale Anki CSV dermatology-relevant cards, filtered for dermatology morphology, infection, malignancy, rash, and systemic-disease skin manifestations.